20261112 Quiz

for (MATH261) Discrete Mathematics

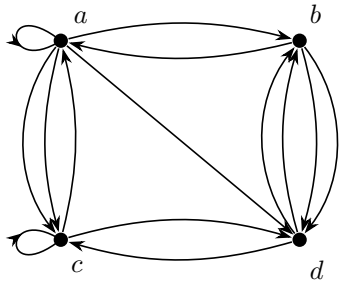
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Problem 1.

(a) Draw the directed graph represented by the given adjacency matrix.

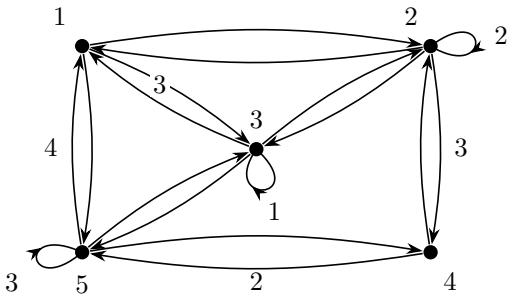
(b) Find the adjacency matrix of the given directed multigraph.

$$\mathbf{A} = \begin{pmatrix} 0 & 1 & 3 & 0 & 4 \\ 1 & 2 & 1 & 3 & 0 \\ 3 & 1 & 1 & 0 & 1 \\ 0 & 3 & 0 & 0 & 2 \\ 4 & 0 & 1 & 2 & 3 \end{pmatrix}.$$



Solution.

(a) Since the matrix is symmetric, its off-diagonal entries give equal numbers of arcs in opposite directions. Each pair of opposite arrows below represents one arc in each direction; a number specifies the multiplicity in *each* direction, while an unlabeled pair has multiplicity one.



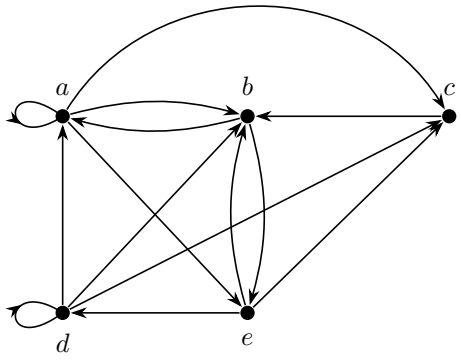
There are 2, 1, and 3 loops at vertices 2, 3, and 5, respectively.

(b) With rows and columns ordered as a, b, c, d, the adjacency matrix is

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 1 & 0 & 1 & 1 \\ 0 & 2 & 1 & 0 \end{pmatrix}.$$

□

Problem 2. Find all paths (starting at vertex a and ending at vertex b or vertex c) in the following directed graph. What are their lengths? Find the circuits starting at vertex a in the following directed graph.



Solution. A path has no repeated vertices. All paths satisfying the stated endpoints, with their lengths in parentheses, are

Ending at b	Ending at c
(a, b) (1)	(a, c) (1)
(a, c, b) (2)	(a, e, c) (2)
(a, e, b) (2)	(a, b, e, c) (3)
(a, e, c, b) (3)	(a, e, d, c) (3)
(a, e, d, b) (3)	(a, b, e, d, c) (4)
(a, e, d, c, b) (4)	

These are exhaustive: from a the first step is toward b, c, or e; subsequent steps must follow the indicated arrows without revisiting a vertex.

A circuit is a closed trail, so vertices may recur but arcs may not. To give *all* circuits without a long list, use the following exact recursive description. The possible heads of outgoing arcs are

Current vertex	a	b	c	d	e
Next vertex	a, b, c, e	a, e	b	a, b, c, d	b, c, d

Start with (a). From the last vertex v, append every allowed w for which $v \to w$ has not yet been used. Record each extension ending at a, but continue extending it as well; stop only when no unused outgoing arc remains. This generates every circuit exactly once, since each closed trail has a unique sequence of such extensions. The loop (a, a) and circuits revisiting a are included. The numbers of circuits of each positive length are

Length	1	2	3	4	5	6	7	8	9	10	Total
Number	1	1	5	11	20	37	55	80	86	36	332

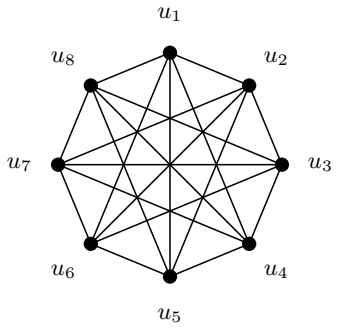
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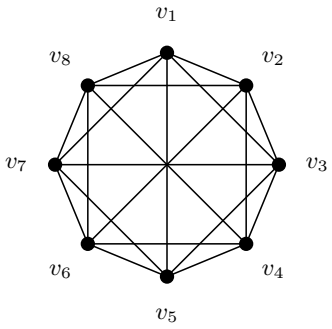
Problem 3. Show that isomorphism of simple graphs is an equivalence relation.

Solution. For every simple graph G, the identity on $V(G)$ is an isomorphism, so the relation is reflexive. If $f\colon V(G) \to V(H)$ is an isomorphism, then $f^{-1}\colon V(H) \to V(G)$ is also one, proving symmetry. Finally, if $f\colon V(G) \to V(H)$ and $g\colon V(H) \to V(K)$ are isomorphisms, so is $g \circ f$, proving transitivity. Thus graph isomorphism is an equivalence relation.

□

Problem 4. Determine whether the given pair of graphs is isomorphic. Exhibit an isomorphism or provide a rigorous argument that none exists.





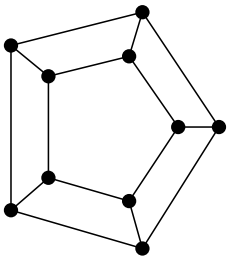
Solution. The graphs are *not* isomorphic. Read the vertex subscripts modulo 8. In the left graph, u_i is adjacent to $u_{i\pm1}$, $u_{i\pm3}$, and u_{i+4}. Hence its complement consists of

$$u_1u_3u_5u_7u_1 \qquad \text{and} \qquad u_2u_4u_6u_8u_2,$$

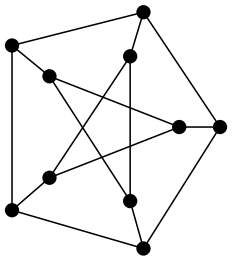
two disjoint 4-cycles. In the right graph, v_i is adjacent to $v_{i\pm1}$, $v_{i\pm2}$, and v_{i+4}, so its complement contains precisely the edges $v_iv_{i\pm3}$. Since $\gcd(3,8) = 1$, these form a single 8-cycle. The complements have different numbers of connected components and therefore cannot be isomorphic. It follows that the original graphs are not isomorphic.

□

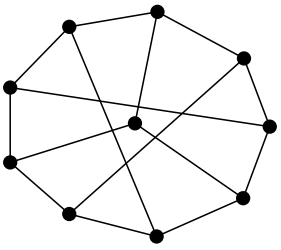
Problem 5. Determine which pairs of the following graphs are isomorphic. For each pair, either exhibit an isomorphism or provide a rigorous argument that none exists.



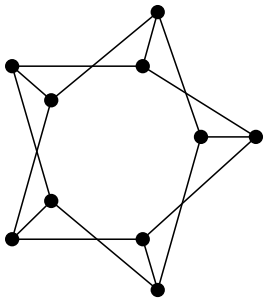
G_1



G_2



G_3



G_4

Solution. Only $G_2 \cong G_3$.

To exhibit an isomorphism, label the outer vertices of G_2 by x_0, x_1, x_2, x_3, x_4, clockwise from the top, and its corresponding inner vertices by y_0, y_1, y_2, y_3, y_4. In G_3, label the nine outer vertices by $z_0, \ldots, z_8$, clockwise from the top, and the central vertex by z_9. The following bijection preserves all 15 edges:

Vertex	x_0	x_1	x_2	x_3	x_4
Image	z_0	z_1	z_2	z_3	z_9
Vertex	y_0	y_1	y_2	y_3	y_4
Image	z_8	z_5	z_7	z_4	z_6

The outer 5-cycle maps to $z_0z_1z_2z_3z_9z_0$. The five spokes map to z_0z_8, z_1z_5, z_2z_7, z_3z_4, and z_9z_6; the inner edges map to z_8z_7, z_8z_4, z_5z_4, z_5z_6, and z_7z_6. These are exactly the edges of G_3, so $G_2 \cong G_3$.

For the remaining pairs, G_1 contains a 4-cycle, while G_2 has none. Indeed, the two 5-cycles of G_2 are joined by a matching; a 4-cycle using two matching edges would require the same pair of indices to be adjacent in both 5-cycles, which is impossible (outer edges join consecutive indices, and inner edges join indices differing by two). Since $G_2 \cong G_3$, neither G_2 nor G_3 is isomorphic to G_1. Moreover, G_4 is bipartite: it consists of a 10-cycle together with five chords joining vertices at distance 5 along that cycle, all of which join opposite parts of the alternating bipartition. By contrast, G_1, G_2, and G_3 each contain a 5-cycle, so none is bipartite. Consequently, G_4 is not isomorphic to any of them.

□